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Investor Sentiment and the Pricing of Characteristics-Based Factors

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1 Big picture

- **Question.** What do characteristics-based factor betas actually measure? Are they exposures to genuine systematic risk, or are they noisy proxies for systematic mispricing?
- **Motivating puzzle.** Earlier work shows that when one sorts firms directly on their exposure to prominent characteristics-based factors, the average return spread between high-beta and low-beta firms is usually tiny or insignificant, even though those factors are very successful at pricing anomaly portfolios.
- **Core idea.** Investor sentiment should move beta-sorted returns in opposite directions depending on what the factor represents. If a factor is a *risk factor*, the beta-return relation should be stronger after *low*-sentiment periods, when investors are relatively more rational and care more about risk. If a factor is a *mispricing factor*, the beta-return relation should be stronger after *high*-sentiment periods, when overpricing is more likely because of short-sale impediments.
- **Empirical design.** The paper studies 13 prominent characteristics-based factors, forms decile portfolios sorted directly on ex ante factor beta, and compares their returns across high- and low-sentiment regimes using the Baker-Wurgler sentiment index. It then repeats the analysis with predictive regressions, mutual funds, hedge funds, alternative sentiment measures, placebo predictors, and overnight versus intraday returns.
- **Main stock-level result.** The unconditional high-minus-low beta spread is close to zero, but a sharp two-regime pattern emerges once sentiment is brought in. Averaging across the 12 characteristics-based factors excluding SMB, the beta-sorted long-short spread is about -0.31% per month after low-sentiment periods and $+0.64\%$ per month after high-sentiment periods, a difference of about 0.95% per month.
- **Main interpretation.** This sign pattern is the opposite of what one gets for macro-related risk factors. The authors therefore argue that exposure to characteristics-based factors is more naturally interpreted as a noisy proxy for the level of mispricing—especially during high-sentiment periods—whereas exposure to macro factors more naturally reflects risk.
- **Cross-sectional asymmetry.** The sentiment effect is much stronger for *low-beta* firms than for high-beta firms. The low-beta side looks especially overpriced after high-sentiment periods, consistent with short-sale constraints making overpricing easier to sustain than underpricing.
- **Mutual fund and hedge fund evidence.** The same two-regime pattern appears when testing assets are mutual funds and hedge funds sorted on their factor betas. This matters because it shows the result is not just about individual stocks.
- **Why this matters.** The paper changes how one should interpret factor “alphas” in anomaly tests, mutual fund performance evaluation, attribution, and event studies. If characteristics-based factor beta is not risk exposure but instead a noisy proxy for mispricing, then adjusting returns by these factor loadings can remove economically meaningful returns that are not true compensation for risk.
- **Economic takeaway.** The paper is not merely saying that sentiment matters. It is saying that *the way sentiment matters is diagnostic of the nature of the factor*. Characteristics-based factors look behavioral or mispricing-related; macro factors look risk-related.

2 Data and measurement (key objects)

2.1 Investor sentiment measures

- **Main sentiment measure.** The baseline measure is the Baker-Wurgler (BW) investor sentiment index.
- **Construction of BW.** It is the first principal component of five sentiment proxies after orthogonalizing them with respect to macroeconomic variables. The ingredients include the average closed-end fund discount, the number of IPOs, average IPO first-day returns, the dividend premium, and the equity share in new issues.
- **Why BW matters here.** The paper needs a market-wide state variable that captures when overpricing is likely to be more severe. The BW index is the canonical measure for that purpose.
- **Alternative sentiment measures for robustness.** The paper also uses the University of Michigan Consumer Sentiment Index, the Conference Board Consumer Confidence Index, the Huang et al. PLS sentiment index, and the AAI sentiment survey index.

2.2 Characteristics-based factors and factor-beta portfolios

- **The 13 factors.** SMB, HML, RMW, CMA, IA, ROE, PMU, QMJ, MOM, MGMT, PERF, FIN, and PEAD.
- **Economic grouping.**
 - Fama-French style factors: SMB, HML, RMW, CMA.
 - Hou-Xue-Zhang style factors: IA and ROE.
 - Other prominent characteristics-based factors: PMU, QMJ, MOM.
 - More explicitly behavioral or mispricing-oriented factors: MGMT, PERF, FIN, and PEAD.
- **Beta estimation windows.**
 - For MOM, PEAD, and PERF, ex ante betas are estimated from daily returns over the past 3 months with a minimum of 45 days.
 - For the other factors, ex ante betas are estimated from monthly returns over the past 60 months with a minimum of 36 months.
- **Why different windows?** The short-horizon factors (especially MOM, PEAD, and PERF) are thought to vary more quickly, so a shorter estimation window is used to maximize the ex post beta spread.
- **Portfolio construction.** For each factor, the authors form 10 value-weighted stock portfolios sorted on ex ante beta and then study the bottom decile (low beta), top decile (high beta), and the high-minus-low spread.

2.3 Stock sample and time span

- The stock sample includes ordinary common shares on NYSE, AMEX, and NASDAQ.

- The main stock sample runs from 1966:7 to 2023:12, with factor-specific starting dates depending on data availability.
- Delisting bias is adjusted following Shumway (1997).

2.4 Mutual fund and hedge fund samples

- **Mutual funds.** Data come from the CRSP Survivor-Bias-Free U.S. Mutual Fund Database and Thomson Reuters Mutual Fund Holdings.
- **Mutual fund filters.** The authors keep actively managed diversified U.S. equity funds, drop balanced/bond/index/international/commodity/sector funds, require at least 70% stock holdings and at least 10 stocks, aggregate across share classes, require lagged fund size above \$15 million (2011 dollars), and drop the first 3 years of returns to mitigate incubation bias.
- **Mutual fund sample size.** 3,566 unique funds and 531,727 fund-month observations from 1980:1 to 2023:12.
- **Hedge funds.** Data come from Thomson Reuters Lipper Hedge Fund Database (TASS), including both live and graveyard funds with returns net of fees.
- **Hedge fund filters.** The paper requires dollar-denominated monthly data, initial NAV above \$10 million, at least 24 months of return history, and drops the first year of returns to mitigate backfill bias.
- **Hedge fund sample size.** 8,233 unique funds and 601,956 fund-month observations from 1996:7 to 2023:12.

2.5 Key empirical objects from Table 1 and Table 2

- **Factor returns and alphas.** Most factor returns and CAPM alphas are significantly positive; SMB is the main exception, consistent with the weak size effect in recent decades.
- **Large ex post beta spreads.** The beta-sorted portfolios deliver economically and statistically large ex post factor-beta spreads. The average ex post spread across the 13 factors is about 1.89 with a t -statistic around 38.64.
- **But weak unconditional pricing.** Despite these large ex post beta spreads, the return spread between high-beta and low-beta firms is generally tiny and usually insignificant.
- **Low correlation between beta and characteristics.** The time-series average of the cross-sectional signed correlation between factor beta and the factor's underlying characteristic is only about 0.15 on average.
- **Low overlap with anomaly portfolios.** Less than 10% of stocks overlap, on average, between the long-short beta-sorted portfolios and the corresponding characteristics-sorted anomaly portfolios.
- **Moderate correlation, not identity.** The return spreads of beta-sorted portfolios and characteristics-sorted portfolios have only moderate correlation, about 0.5 on average.
- **Interpretation.** This is one of the paper's most important measurement points: a high-beta stock is *not* just the same thing as a stock with an extreme characteristic. That is why the

paper’s tests do not simply restate the known result that anomaly profits are stronger after high sentiment.

2.6 What is observed and what is not

- The paper observes returns, fund returns, factor returns, market shares, and lagged sentiment.
- It does *not* observe the true latent “mispricing” level. Factor beta is treated as a noisy proxy for that latent object.
- This is why the paper’s central identification device is sign-based and state-dependent: the same beta-sorted portfolios should react differently to sentiment depending on whether beta measures risk or mispricing.

3 Models and empirical specifications (all equations + interpretation)

3.1 Ex ante beta estimation

For each stock or fund, the paper estimates factor beta from a single-factor regression:

$$r_{i,\tau} = \alpha_i^f + \beta_i^f F_\tau^f + \varepsilon_{i,\tau},$$

where F_τ^f is one of the 13 characteristics-based factors.

Interpretation. The object of interest is not the characteristic itself, but the estimated exposure to the factor. The entire paper then asks whether higher estimated exposure is rewarded differently after high versus low sentiment.

3.2 Two-regime portfolio design

Let S_{t-1} denote lagged sentiment. A month is classified as *high sentiment* if S_{t-1} is in the top 50% of the sentiment series and *low sentiment* if it is in the bottom 50%.

For each factor f , the paper computes average returns for:

- the low-beta portfolio,
- the high-beta portfolio,
- the high-minus-low spread.

Interpretation. This is the simplest and most transparent test. If factor beta proxies for risk, high-minus-low should be more positive after *low* sentiment. If factor beta proxies for underpricing/mispricing, high-minus-low should be more positive after *high* sentiment.

3.3 Predictive regressions with lagged sentiment

The next specification is

$$R_t^p = a_p + b_p S_{t-1} + e_t^p,$$

where R_t^p is the excess return on a beta-sorted portfolio.

The paper also estimates

$$R_t^p = a_p + b_p S_{t-1} + c_p MKT_t + e_t^p,$$

and versions that add standard macro predictors.

Interpretation.

- For a high-minus-low beta spread, a *positive* b_p supports the mispricing view.
- For the low-beta portfolio, a *negative* b_p means low-beta firms earn especially low future returns after high sentiment, consistent with stronger overpricing on the low-beta side.
- For the high-beta portfolio, a weak or insignificant b_p means the long leg is much less sentiment-sensitive.

3.4 Predictive regressions with additional macro controls

The extended regression is of the form

$$R_t^p = a_p + b_p S_{t-1} + \Gamma_p' X_{t-1} + e_t^p,$$

where X_{t-1} includes variables such as the default premium, the term premium, the real interest rate, and the log wealth-consumption ratio, and in later robustness tables also TED, inflation, and aggregate dispersion.

Interpretation. This specification checks whether the sentiment effect is really just a repackaging of standard macro or risk-premium predictors. The answer in the paper is no.

3.5 Contemporaneous sentiment-change regressions

To study why the unconditional beta-return relation is weak, the paper runs

$$R_t^p = a_p + b_p \Delta S_t + c_p MKT_t + e_t^p.$$

Interpretation. If low-beta firms are more exposed to sentiment shocks, then they should co-move more strongly with contemporaneous changes in sentiment. This helps explain why the unconditional expected-return spread can be flat even when the conditional two-regime pattern is strong.

3.6 Fund-level analogs

The same sort of beta estimation and portfolio sorting is repeated for mutual funds and hedge funds:

- sort funds on their factor betas,
- form low-beta and high-beta fund portfolios,
- compare their average returns across sentiment states.

Interpretation. This is a powerful external check. If the mechanism is truly about what the factor beta captures, it should show up in professionally managed portfolios as well, not just in stock-level sorts.

3.7 Overnight versus intraday decomposition

Following Lou, Polk, and Skouras (2019), the paper splits daily returns into intraday and overnight components and repeats the beta-sorted pricing tests separately.

Interpretation. The authors use this as another way to distinguish more rational from less rational trading environments:

- overnight is interpreted as relatively more rational,
- intraday as relatively less rational.

If characteristics-based betas proxy for mispricing, they should behave more like the *high-sentiment* regime intraday and more like the *low-sentiment* regime overnight.

3.8 What the paper is not doing

- It is not estimating a structural model of sentiment.
- It is not instrumenting sentiment in the standard IV sense.
- It is not claiming that characteristics-based factors are *purely* mispricing. The paper repeatedly allows for the possibility that they combine both risk and mispricing, but shows that the sign pattern lines up much more naturally with the mispricing view.

4 Identification and instruments (what makes the estimates credible)

4.1 Main identification logic

The paper's identification is not based on a conventional instrument. It is based on a *comparative sign test*:

- **Risk-factor prediction:** high-minus-low beta spreads should be larger after low-sentiment periods.
- **Mispricing-factor prediction:** high-minus-low beta spreads should be larger after high-sentiment periods.

Because these predictions have opposite signs, the effect of sentiment on beta-sorted spreads is informative about the underlying force behind the factor.

4.2 Why beta-sorted portfolios are the right testing assets

- The traditional anomaly portfolios are built directly from characteristics. If anomalies themselves are stronger after high sentiment, that fact does not distinguish risk from mispricing in the factor.
- By contrast, the paper sorts on *factor beta*. Since factor beta is only weakly correlated with the underlying characteristic (about 15% on average) and overlaps little with the anomaly portfolios (about 10% overlap), the beta-sorted portfolios provide a genuinely different testing laboratory.

- The large ex post beta spreads make the testing assets powerful. This addresses the concern that insignificant unconditional spreads merely reflect weak sorts.

4.3 Why the exercise is credible

- **Opposite benchmark from macro factors.** The same methodology yields the opposite sign for macro-related factors, exactly as earlier work on risk-based macro factors suggests. That is a strong discipline check.
- **Robust across asset classes.** The two-regime pattern appears in stocks, mutual funds, and hedge funds.
- **Robust across sentiment measures.** BW, Michigan, Conference Board, PLS, and AAI all give the same qualitative message.
- **Robust to controls.** The predictive relation survives adding the market factor, macro variables, TED, inflation, aggregate dispersion, and one-by-one equity premium predictors.
- **Robust to beta estimation windows.** Shorter and longer beta windows, as well as two-factor beta estimation with the market factor, leave the main message intact.
- **Robust to small-stock concerns.** Using NYSE breakpoints, excluding microcaps, and excluding penny stocks do not materially change the results.
- **Robust to removing explicit behavioral factors.** Excluding MGMT, PERF, FIN, and PEAD still leaves the same broad pattern.

4.4 Why the results are not a mechanical consequence of anomaly timing

The paper makes a subtle but important point. Stambaugh, Yu, and Yuan (2012) show that anomaly spreads are stronger after high sentiment. One might think this automatically implies that beta-sorted portfolios should also do better after high sentiment. But that conclusion would only be mechanical if beta and characteristics were nearly the same object. They are not:

- beta-characteristic correlation is only about 0.15 on average,
- stock overlap between beta-sorted and characteristics-sorted long-short portfolios is only about 10% on average.

So the two-regime beta result contains new information.

4.5 What makes the sentiment story especially persuasive

The strongest persuasive feature is the *asymmetry*:

- low-beta firms are hit hard after high sentiment,
- high-beta firms move much less,
- most of the alpha in the high-minus-low spread comes from the short leg in high-sentiment periods.

Table 1
Summary statistics of the characteristics-based factors
A. Pairwise correlations among characteristics-based factors

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
(1) SMB	1.00												
(2) HML	-.01	1.00											
(3) RMW	-.36	.08	1.00										
(4) CMA	-.09	.09	-.01	1.00									
(5) IA	-.13	.68	.09	.82	1.00								
(6) ROE	-.40	-.12	.66	-.05	.05	1.00							
(7) PMU	-.02	-.39	.45	-.36	-.32	.39	1.00						
(8) QMT	-.49	-.03	.69	.09	.14	.70	.46	1.00					
(9) MCM	-.07	-.19	.09	-.01	.04	.49	.07	.30	1.00				
(10) MGMT	-.29	.66	.23	.78	.75	.09	-.18	.38	-.05	1.00			
(11) FBRE	-.27	-.35	.53	-.11	-.07	.72	.48	.69	.64	-.02	1.00		
(12) FIN	-.38	.61	.59	.59	.63	.37	.02	.55	.09	.74	.22	1.00	
(13) PEAD	-.06	-.20	-.08	-.05	-.08	-.24	.09	.18	.47	-.05	.33	-.04	1.00

B. Factor returns and CAPM alphas

Return	0.21 (1.80)	0.29 (2.18)	0.28 (3.13)	0.27 (3.10)	0.36 (4.10)	0.53 (5.28)	0.33 (3.39)	0.38 (4.03)	0.60 (3.80)	0.51 (5.13)	0.56 (5.03)	0.71 (4.26)	0.57 (7.37)
Alpha ^{CAPM}	0.11 (0.93)	0.37 (2.60)	0.34 (3.54)	0.37 (4.28)	0.45 (5.23)	0.61 (6.59)	0.32 (3.09)	0.53 (6.54)	0.69 (4.79)	0.65 (6.94)	0.64 (6.29)	0.95 (6.15)	0.61 (8.09)

C. Correlations between characteristics-based factors and BW sentiment index

$\bar{S}_{t-1}$	-.11	.06	.15	.08	.07	.09	.12	.18	.01	.11	.09	.10	-.03
$\bar{S}_t$	.09	-.15	-.12	-.10	-.08	-.04	.09	-.06	.03	-.14	.00	-.17	.03

This table reports the summary statistics of characteristics-based tradable factors. Panel A presents the pairwise Pearson correlations of the 13 factors' monthly returns. Panel B presents the factors' monthly mean returns (in percentages) and CAPM alphas (in percentages) with Newey and West (1987) five-lag adjusted *t*-statistics in parentheses. Panel C presents the correlations between the Baker Wurgler (BW) sentiment index and the characteristics-based factors. The sample period is from 1967:1 to 2023:12 for IA and ROE factors, 1972:7 to 2023:12 for FIN and PEAD factors, and 1967:7 to 2023:12 for other factors. The sample period is 1965:07 to 2023:12 for the BW sentiment index.

That is exactly what one expects if sentiment mainly creates overpricing that is hard to arbitrage away because of short-sale constraints.

4.6 Main limitations

- **Indirect inference.** The paper does not observe mispricing directly; it infers it from how beta-sorted returns move with sentiment.
- **Sentiment may still bundle multiple forces.** Even after many controls, sentiment could capture a mix of disagreement, extrapolation, funding constraints, and other behavioral channels.
- **Beta is only a noisy proxy.** The paper explicitly emphasizes that factor beta may be a noisy proxy for underpricing, not a clean one.
- **No clean decomposition into “pure risk” and “pure mispricing.”** The results are strongest for saying that mispricing plays at least an important role, not that risk plays none.
- **Mostly U.S.-based evidence.** External validity to other equity markets is not established here.

4.7 Relation to the literature

- Baker and Wurgler (2006, 2007): sentiment matters more for hard-to-arbitrage, hard-to-value securities.
- Stambaugh, Yu, and Yuan (2012): anomalies are stronger after high sentiment.
- Shen, Yu, and Zhao (2017): macro risk-factor beta spreads are stronger after low sentiment.
- Chen et al. (2018) and Daniel et al. (2020): characteristics-based factors are not priced when testing assets are beta-sorted portfolios.

How this paper fits in. It combines these literatures and turns the sentiment state into a diagnostic tool for factor interpretation.

5 Tables and figures

5.1 Table 1: characteristics-based factor summary statistics

Read: This table tells you what the 13 factors look like before any beta sorts are formed.

- Panel A reports pairwise factor correlations. The factors are related, but not identical.
- Panel B reports mean returns and CAPM alphas. Almost all factors earn positive raw returns and positive CAPM alphas; SMB is the main weak exception.
- Panel C reports correlations with lagged BW sentiment and contemporaneous changes in sentiment. Most factors are only weakly related to lagged sentiment.

Why it matters. The paper is not just cherry-picking weird factors with no return content. These are the canonical characteristics-based factors that have been widely used in the anomaly and factor-pricing literature.

5.2 Table 2: diagnostics for beta-sorted portfolios

Table 2
Beta-sorted portfolios

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
<i>A. Pairwise correlations among beta-sorted high-minus-low portfolios</i>															
(1) SMB	1.00														
(2) HML	-.31	1.00													
(3) RMW	-.46	.32	1.00												
(4) CMA	-.43	.81	.33	1.00											
(5) IA	-.45	.83	.40	.93	1.00										
(6) ROE	-.75	.12	.80	.19	.25	1.00									
(7) PMU	-.86	-.31	.19	-.36	-.30	.28	1.00								
(8) QMJ	-.87	.29	.74	.44	.46	.80	.19	1.00							
(9) MOM	-.82	-.09	.11	.01	.01	.15	.06	.14	1.00						
(10) MOMT	-.63	.75	.69	.87	.87	.36	-.18	.64	.00	1.00					
(11) PERF	-.34	-.02	.48	.09	.12	.55	.18	.46	.60	.15	1.00				
(12) FSN	-.48	.69	.67	.76	.80	.50	.04	.71	.03	.88	.28	1.00			
(13) PEAD	-.04	-.18	-.01	-.09	-.13	.06	.03	.08	.39	-.07	.24	-.08	1.00		
(14) Ave	-.63	.61	.75	.70	.74	.66	.30	.77	.37	.79	.59	.85	.18	1.00	
(15) $Ave_{high-low}$	-.72	.59	.76	.70	.74	.70	.09	.83	.34	.80	.58	.87	.16	.99	1.00
<i>B. Excess returns and CAPM alphas</i>															
Ref_{β}^H	0.38	0.81	0.54	0.52	0.53	0.56	0.67	0.54	0.49	0.60	0.68	0.70	0.65	0.57	0.59
Ref_{β}^L	(1.82)	(1.23)	(2.62)	(2.43)	(2.54)	(2.85)	(2.44)	(3.60)	(3.84)	(3.34)	(3.07)	(3.75)	(2.21)	(2.92)	(3.19)
Ref_{β}^L	0.56	0.57	0.23	0.58	0.63	0.64	0.38	0.39	0.47	0.56	-.015	0.46	0.49	0.42	0.41
Ref_{β}^L	(3.92)	(1.75)	(0.69)	(1.74)	(1.76)	(1.81)	(1.38)	(1.02)	(1.55)	(1.50)	(-0.45)	(1.20)	(1.50)	(1.40)	(1.36)
Ref_{β}^L	-.017	0.24	0.31	-.006	-.010	-.009	0.29	0.15	0.01	0.04	0.83	0.24	0.16	0.15	0.18
Ref_{β}^L	(-0.57)	(0.78)	(1.07)	(-0.19)	(-0.30)	(-0.29)	(1.01)	(0.46)	(0.06)	(0.11)	(3.04)	(0.70)	(0.56)	(0.97)	(0.96)
$Alpha_{\beta}^{APM}$	-0.71	0.48	0.62	0.28	0.26	0.28	0.25	0.73	0.11	0.55	1.06	0.85	0.18	0.37	0.46
$Alpha_{\beta}^{APM}$	(-2.93)	(1.44)	(2.30)	(0.96)	(0.77)	(0.98)	(0.83)	(2.74)	(0.37)	(1.81)	(3.92)	(2.64)	(0.59)	(2.61)	(2.84)

(Continued)

Table 2
(Continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
<i>C. Ex post betas</i>															
High beta	1.97	0.56	0.23	0.01	0.13	0.07	0.92	-0.29	0.36	-0.18	[0.21]	-0.05	0.28	0.37	0.21
	(16.55)	(3.59)	(1.96)	(0.06)	(0.96)	(0.69)	(8.01)	(-3.25)	(2.66)	(-1.72)	(1.77)	(-0.63)	(1.03)	(7.51)	(4.18)
Low beta	-0.01	-1.33	-2.10	-2.09	-2.17	-1.79	-1.13	-2.89	-1.08	-2.51	-1.67	-1.71	-1.09	-1.52	-1.67
	(-0.10)	(-7.68)	(-10.71)	(-11.36)	(-9.25)	(-9.78)	(-6.73)	(-19.84)	(-9.33)	(-14.28)	(-11.51)	(-17.61)	(-4.23)	(-25.33)	(-26.45)
High - Low	1.98	1.88	2.32	2.10	2.30	1.85	2.05	2.60	1.44	2.32	1.88	1.67	1.37	1.89	1.88
	(16.65)	(14.66)	(17.54)	(11.90)	(12.69)	(20.29)	(20.60)	(15.44)	(15.32)	(18.86)	(23.39)	(8.91)	(18.64)	(37.07)	
<i>D. Cross-sectional signed correlations: Beta versus characteristics</i>															
Corr	.17	.20	.05	.07	.05	.07	.18	.36	.22	.13	.18	.28	.02	.15	.15
<i>E. Percentage of overlap stocks in top and bottom decile beta and characteristics portfolio</i>															
% of overlap	13.18	10.57	9.52	8.51	8.14	7.48	8.87	11.10	13.61	6.10	10.12	9.09	8.22	9.58	9.28
<i>F. Correlation of high-minus-low return spreads of beta vs. characteristics portfolios</i>															
Corr	.53	.46	.63	.28	.23	.62	.36	.80	.71	.53	.60	.55	.25	.50	.50

This table reports the results of decile value-weighted portfolios sorted by factor betas. The betas of MOM, PEAD, and PERF are estimated from a single characteristics-based factor model using daily returns over the past 3 months with a minimum of 45 days. The betas of other factors are estimated from a single-factor model using monthly returns over the past 60 months with a minimum of 36 months. All portfolios are rebalanced monthly. Panel A presents the pairwise Pearson correlations of monthly returns for the high-minus-low beta-sorted portfolios. Panel B (C) presents the monthly mean excess returns and CAPM alphas in percentage (ex post betas) of high-beta, low-beta, and high-minus-low portfolios with Newey-West (1987) five-lag adjusted t -statistics in parentheses. Ex post betas are estimated using the corresponding single characteristics-based factor model. Panel D presents the time-series average of cross-sectional signed correlations between factor beta and respective stock characteristics, where the correlation sign is adjusted so that the value of characteristics predicts cross-sectional returns with a positive sign. Panel E presents the time-series average of the percentage of overlap stocks in the top and bottom decile beta and characteristics portfolios. For top (bottom) decile portfolios, we first calculate the percentage of overlap stocks in the top (bottom) decile beta and characteristics portfolios each month. We take an average of the percentages between the top and bottom decile portfolios and then the time-series average of the percentages. Panel F presents the time-series correlations between the return spreads of high-minus-low portfolios for the factor-beta-sorted portfolio and corresponding stock-characteristics sorted portfolio. Ave refers to the average portfolio across all beta-sorted portfolios. Ave ($Ave_{high-low}$) refers to the average portfolio of all beta-sorted portfolios (including SMB). The sample period is from 1970:1 to 2023:12 for IA and ROE beta-sorted portfolios, 1975:1 to 2023:12 for FSN and PEAD beta-sorted portfolios, and 1966:7 to 2023:12 for other beta-sorted portfolios.

Read: This is the table that shows the beta sorts are statistically powerful but economically different from anomaly sorts.

- The ex post beta spreads are large: the average is about 1.89 with a very large t -statistic.
- Yet the unconditional return spreads are small or insignificant.
- The average signed cross-sectional correlation between factor beta and the underlying characteristic is only about 0.15.
- The overlap between beta-sorted and characteristic-sorted portfolios is below 10% on average.
- The correlation between the return spreads of the two kinds of portfolios is only moderate, about 0.5.

Why it matters. Table 2 sets up the whole paper. It says: the beta-sorted portfolios are strong enough to test pricing, but different enough from the anomaly portfolios that the results are not tautological.

5.3 Table 3: the main two-regime stock-level evidence

Read: This is the centerpiece table.

- Averaging across the 12 factors excluding SMB, the high-minus-low beta spread is about -0.31% per month after low sentiment and $+0.64\%$ per month after high sentiment.
- The difference is about 0.95% per month, economically large and statistically significant.

Table 3
Beta-sorted portfolio returns following high and low sentiment

	Low beta			High beta			H-L		
	Low Sent.	High Sent.	High - Low	Low Sent.	High Sent.	High - Low	Low Sent.	High Sent.	High - Low
SMB	0.60 (2.94)	0.52 (2.52)	-0.08 (-0.29)	1.23 (2.29)	-0.44 (-0.89)	-1.67 (-2.33)	0.63 (1.48)	-0.95 (-2.41)	-1.59 (-2.80)
HML	1.06 (2.48)	0.09 (0.20)	-0.97 (-1.57)	0.80 (1.94)	0.82 (2.67)	0.03 (0.05)	-0.26 (-0.64)	0.73 (1.56)	0.99 (1.59)
RMW	1.00 (2.25)	-0.52 (-1.13)	-1.53 (-2.41)	0.53 (1.83)	0.54 (1.87)	0.01 (0.02)	-0.47 (-1.18)	1.06 (2.75)	1.54 (2.77)
CMA	0.95 (2.07)	0.22 (0.48)	-0.73 (-1.14)	0.51 (1.69)	0.53 (1.76)	0.02 (0.04)	-0.44 (-1.10)	0.31 (0.67)	0.74 (1.27)
IA	1.17 (2.35)	0.12 (0.24)	-1.05 (-1.53)	0.53 (1.47)	0.53 (1.72)	0.00 (0.00)	-0.64 (-1.50)	0.41 (0.83)	1.05 (1.65)
ROE	1.25 (2.32)	0.07 (0.14)	-1.18 (-1.67)	0.74 (2.67)	0.38 (1.37)	-0.36 (-0.91)	-0.51 (-1.12)	0.31 (0.77)	0.82 (1.35)
PMU	0.86 (2.24)	-0.08 (-0.20)	-0.94 (-1.71)	0.90 (2.29)	0.45 (1.19)	-0.46 (-0.86)	0.05 (0.12)	0.53 (1.29)	0.48 (0.85)
QMJ	1.29 (2.41)	-0.47 (-0.91)	-1.76 (-2.39)	0.50 (2.31)	0.59 (2.75)	0.09 (0.31)	-0.79 (-1.77)	1.06 (2.50)	1.85 (3.02)
MOM	0.89 (1.99)	0.07 (0.16)	-0.83 (-1.35)	0.85 (2.46)	0.13 (0.39)	-0.72 (-1.48)	-0.04 (-0.11)	0.06 (0.14)	0.10 (0.18)
MGMT	1.24 (2.42)	-0.09 (-0.17)	-1.33 (-1.84)	0.47 (1.83)	0.73 (2.84)	0.25 (0.68)	-0.77 (-1.72)	0.81 (1.70)	1.58 (2.46)
PERF	0.54 (1.18)	-0.83 (-1.66)	-1.37 (-2.05)	1.06 (3.33)	0.31 (1.04)	-0.75 (-1.67)	0.52 (1.43)	1.14 (2.77)	0.63 (1.11)
FIN	1.46 (2.89)	-0.37 (-0.69)	-1.83 (-2.61)	0.80 (2.66)	0.62 (2.59)	-0.19 (-0.48)	-0.65 (-1.46)	0.99 (2.06)	1.64 (2.63)
PEAD	1.11 (2.33)	-0.03 (-0.08)	-1.15 (-1.87)	1.16 (2.88)	0.22 (0.55)	-0.94 (-1.74)	0.05 (0.12)	0.26 (0.62)	0.20 (0.35)
Ave	0.98 (2.43)	-0.12 (-0.31)	-1.10 (-2.00)	0.75 (2.59)	0.39 (1.51)	-0.36 (-0.93)	-0.23 (-1.16)	0.51 (2.33)	0.75 (2.54)
$Ave_{w/o SMB}$	1.02 (2.38)	-0.18 (-0.42)	-1.19 (-2.05)	0.71 (2.60)	0.46 (1.89)	-0.25 (-0.68)	-0.31 (-1.24)	0.64 (2.49)	0.95 (2.72)
Ave^{CAPM}	-0.06 (-0.37)	-0.69 (-4.13)	-0.63 (-2.82)	-0.05 (-0.71)	0.00 (0.02)	0.05 (0.46)	0.01 (0.03)	0.69 (3.37)	0.68 (2.62)
$Ave_{w/o SMB}^{CAPM}$	-0.06 (-0.37)	-0.77 (-4.18)	-0.70 (-2.83)	-0.05 (-0.66)	0.10 (1.10)	0.15 (1.24)	0.01 (0.08)	0.86 (3.74)	0.85 (2.81)

This table reports the results of factor beta-sorted decile value-weighted portfolios following high- and low-sentiment regimes, as classified based on the median level of the Baker and Wurgler (2006) sentiment index. We report the excess returns for the bottom decile portfolio (Low beta), the top decile portfolio (High beta), and their differences (H-L). Preranking betas of MOM, PEAD, and PERF are estimated from a single characteristics-based factor model using daily returns over the past 3 months with a minimum of 45 days. Other preranking betas are estimated from a single-factor model using monthly returns over the past 60 months with a minimum of 36 months. All portfolios are rebalanced monthly. Ave ($Ave_{w/o SMB}$) refers to the average portfolios of all beta-sorted portfolios (excluding SMB). We report CAPM alphas for the two average portfolios (Ave^{CAPM} and $Ave_{w/o SMB}^{CAPM}$) in the last four rows. The sample period is from 1970:1 to 2023:12 for IA and ROE, 1975:7 to 2023:12 for FIN and PEAD, and 1966:7 to 2023:12 for others. Returns and alphas are in percentages. Newey-West (1987) five-lag adjusted *t*-statistics are reported in parentheses.

- Including SMB lowers the average difference, but the pattern remains positive.
- The asymmetry comes mainly from the low-beta side. The paper emphasizes that the low-beta portfolio's return difference across high and low sentiment is about 1.19% per month, while the corresponding difference for the high-beta portfolio is only about 0.25%.

CAPM alpha version.

- After CAPM adjustment, the two-regime pattern remains.
- Following high-sentiment periods, the average CAPM alpha spread is about 0.86% per month.
- Following low-sentiment periods, the average CAPM alpha spread is only about 0.01%.
- Most of this alpha comes from the short leg: the low-beta side has about -0.77% alpha in high-sentiment periods, while the high-beta side is only about +0.10%.

Why it matters. This is exactly what the mispricing interpretation predicts and exactly what the risk interpretation does *not* predict.

5.4 Table 4 and Table 5: predictive regressions

Read: These tables translate the simple two-regime comparisons into formal predictive regressions.

- In Table 4, the high-minus-low spread loads positively on lagged sentiment.

Table 4
Beta-sorted portfolios: Predictive regressions on lagged sentiment
A. Lagged BW sentiment

	Low beta		High beta		H-L	
	$\hat{b}$	t -stat.	$\hat{b}$	t -stat.	$\hat{b}$	t -stat.
SMB	-0.04	(-0.23)	-1.02	(-2.48)	-0.98	(-3.22)
HML	-0.61	(-1.65)	-0.21	(-0.79)	0.40	(1.17)
RMW	-0.96	(-2.87)	-0.07	(-0.26)	0.89	(3.18)
CMA	-0.60	(-1.58)	-0.09	(-0.41)	0.52	(1.69)
IA	-0.82	(-1.79)	0.09	(0.44)	0.91	(2.40)
ROE	-0.81	(-1.96)	-0.11	(-0.41)	0.71	(2.21)
PMU	-0.54	(-2.04)	-0.48	(-1.37)	0.06	(0.19)
QMJ	-1.25	(-3.14)	0.06	(0.32)	1.31	(4.53)
MOM	-0.64	(-1.96)	-0.50	(-1.92)	0.14	(0.53)
MGMT	-1.05	(-2.53)	-0.01	(-0.07)	1.04	(3.02)
PERF	-1.24	(-3.26)	-0.22	(-0.87)	1.03	(3.88)
FIN	-1.39	(-3.00)	-0.07	(-0.37)	1.32	(3.21)
PEAD	-0.65	(-1.95)	-0.87	(-2.29)	-0.23	(-0.54)
Ave	-0.79	(-2.50)	-0.28	(-1.21)	0.51	(3.46)
Ave _{w/o SMB}	-0.85	(-2.59)	-0.21	(-0.98)	0.64	(3.67)

B. Lagged BW sentiment, controlling market factor

	Low beta		High beta		H-L	
	$\hat{b}$	t -stat.	$\hat{b}$	t -stat.	$\hat{b}$	t -stat.
SMB	0.15	(2.56)	-0.59	(-2.96)	-0.74	(-3.15)
HML	-0.24	(-1.29)	0.06	(0.35)	0.30	(0.94)
RMW	-0.59	(-3.01)	0.17	(1.30)	0.75	(2.75)
CMA	-0.22	(-1.41)	0.14	(1.09)	0.36	(1.47)
IA	-0.53	(-2.89)	0.27	(1.81)	0.80	(2.69)
ROE	-0.53	(-2.43)	0.06	(0.51)	0.59	(2.05)
PMU	-0.24	(-1.16)	-0.16	(-1.08)	0.08	(0.26)
QMJ	-0.81	(-4.18)	0.25	(3.00)	1.06	(4.40)
MOM	-0.30	(-1.73)	-0.21	(-1.61)	0.09	(0.38)
MGMT	-0.63	(-3.62)	0.18	(1.60)	0.81	(3.13)
PERF	-0.88	(-4.47)	0.05	(0.48)	0.93	(3.89)
FIN	-0.92	(-3.85)	0.14	(1.11)	1.07	(3.33)
PEAD	-0.29	(-1.14)	-0.52	(-2.53)	-0.23	(-0.57)
Ave	-0.42	(-4.06)	-0.01	(-0.24)	0.41	(3.36)
Ave _{w/o SMB}	-0.48	(-4.10)	0.04	(0.68)	0.51	(3.70)

This table reports the point estimates of b ($\hat{b}$), along with Newey-West (1987) five-lag adjusted t -statistics, in the regression specifications: $R_{i,t} = a + bS_{t-1} + e_{i,t}$ (panel A) and $R_{i,t} = a + bS_{t-1} + cMKTr_t + e_{i,t}$ (panel B). Excess returns ($R_{i,t}$) are in percentages, and the sentiment index is scaled to have a zero mean and unit standard deviation. Point estimates of b are presented for the bottom decile portfolio (Low beta), the top decile (High beta) portfolio, and their differences (H-L). Preranking betas of MOM, PEAD, and PERF are estimated from a single characteristics-based factor model using daily returns over the past 3 months with a minimum of 45 days. Other preranking betas are estimated from a single factor model using monthly returns over the past 60 months with a minimum of 36 months. All portfolios are rebalanced monthly. Ave (Ave_{w/o SMB}) refers to the average portfolios of all beta-sorted portfolios (excluding SMB). The sample period is from 1970:1 to 2023:12 for IA and ROE, 1975:7 to 2023:12 for FIN and PEAD, and 1966:7 to 2023:12 for others.

- For the average portfolio excluding SMB, a one-standard-deviation increase in lagged sentiment predicts about a 0.64% higher monthly return on the long-short beta strategy.
- The low-beta leg has a strongly negative coefficient; the high-beta leg has a much weaker coefficient.
- Adding the market factor leaves the main pattern intact.

Table 5 then adds standard macro variables such as the default premium, term premium, real interest rate, and wealth-consumption ratio.

- The sentiment coefficients barely move.
- For the average portfolio excluding SMB, the high-minus-low coefficient remains about 0.61% per month and highly significant.

Why it matters. The sentiment effect is not simply a disguised business-cycle or risk-premium predictor.

5.5 Table 6: contemporaneous sentiment changes

Read: This table helps explain why the unconditional beta-return relation is flat.

- Low-beta portfolios co-move more strongly with contemporaneous sentiment changes.

Table 5
Beta-sorted portfolios: Predictive regressions on lagged sentiment and macro variables

	Low beta		High beta		H-L	
	$\hat{b}$	t -stat.	$\hat{b}$	t -stat.	$\hat{b}$	t -stat.
SMB	-0.02	(-0.11)	-0.84	(-2.12)	-0.82	(-2.78)
HML	-0.56	(-1.52)	-0.13	(-0.51)	0.43	(1.24)
RMW	-0.89	(-2.71)	0.00	(0.00)	0.89	(3.22)
CMA	-0.52	(-1.34)	-0.03	(-0.15)	0.49	(1.52)
IA	-0.75	(-1.65)	0.17	(0.84)	0.92	(2.29)
ROE	-0.66	(-1.67)	-0.10	(-0.38)	0.56	(1.87)
PMU	-0.44	(-1.68)	-0.39	(-1.13)	0.05	(0.16)
QMJ	-1.07	(-2.77)	0.08	(0.44)	1.15	(4.04)
MOM	-0.54	(-1.64)	-0.46	(-1.78)	0.09	(0.33)
MGMT	-0.96	(-2.30)	0.03	(0.17)	1.00	(2.85)
PERF	-1.15	(-2.99)	-0.17	(-0.69)	0.98	(3.62)
FIN	-1.40	(-2.97)	-0.02	(-0.09)	1.39	(3.29)
PEAD	-0.55	(-1.66)	-0.85	(-2.20)	-0.29	(-0.72)
Ave	-0.70	(-2.25)	-0.21	(-0.95)	0.49	(3.19)
Ave _{w/o SMB}	-0.76	(-2.33)	-0.16	(-0.74)	0.61	(3.38)

This table reports the point estimates of b ($\hat{b}$), along with Newey-West (1987) five-lag adjusted t -statistics, in the regression specifications: $R_{i,t} = a + b \times S_{t-1} + \sum_{j=1}^4 m_j \times X_{j,t-1}$. Macro variables $X_{j,t-1}$ include the default premium, the term premium, the real interest rate, and the log wealth-consumption ratio. Excess returns ($R_{i,t}$) are in percentages, and the sentiment index is scaled to have a zero mean and unit standard deviation. Point estimates of b are presented for the bottom decile portfolio (Low beta), the top decile portfolio (High beta), and their differences (H-L). Preranking betas of MOM, PEAD, and PERF are estimated from a single characteristics-based factor model using daily returns over the past 3 months with a minimum of 45 days. Other preranking betas are estimated from a single-factor model using monthly returns over the past 60 months with a minimum of 36 months. All portfolios are rebalanced monthly. Ave (Ave_{w/o SMB}) refers to the average portfolios of all beta-sorted portfolios (excluding SMB). The sample period is from 1970:1 to 2023:12 for IA and ROE, 1975:7 to 2023:12 for FIN and PEAD, and 1966:7 to 2023:12 for others.

- For the average portfolio excluding SMB, the coefficient is about $+0.32$ for the low-beta leg and about -0.16 for the high-beta leg, so the high-minus-low spread has a coefficient around -0.48 .

Interpretation. Low-beta firms are more sentiment-sensitive. When sentiment is high, they become relatively more overpriced; when sentiment is low, they become relatively less overpriced or slightly underpriced. Combining these regimes flattens the unconditional beta-return relation.

Table 6
Beta-sorted portfolios: Contemporaneous regressions on change in sentiment

	Low beta		High beta		H-L	
	$\hat{b}$	t -stat.	$\hat{b}$	t -stat.	$\hat{b}$	t -stat.
SMB	-0.18	(-2.32)	0.45	(1.63)	0.63	(1.88)
HML	0.45	(2.06)	-0.51	(-1.95)	-0.96	(-2.22)
RMW	0.47	(1.86)	-0.23	(-1.64)	-0.70	(-2.03)
CMA	0.52	(2.67)	-0.44	(-1.46)	-0.97	(-2.27)
IA	0.49	(2.27)	-0.47	(-1.53)	-0.96	(-2.21)
ROE	0.23	(0.82)	-0.16	(-1.22)	-0.39	(-1.04)
PMU	-0.25	(-0.80)	0.28	(1.79)	0.53	(1.38)
QMJ	0.42	(1.51)	-0.19	(-1.92)	-0.61	(-1.76)
MOM	-0.12	(-0.40)	0.27	(1.35)	0.39	(0.91)
MGMT	0.54	(2.40)	-0.46	(-2.68)	-0.99	(-2.82)
PERF	0.40	(1.41)	-0.10	(-0.63)	-0.50	(-1.32)
FIN	0.71	(2.80)	-0.32	(-2.26)	-1.04	(-3.00)
PEAD	0.05	(0.23)	0.34	(1.54)	0.29	(0.87)
Ave	0.28	(1.81)	-0.11	(-1.20)	-0.39	(-2.45)
Ave _{wo} /o SMB	0.32	(1.86)	-0.16	(-1.78)	-0.48	(-2.53)

This table reports the point estimates of b ($\hat{b}$), along with Newey-West (1987) five-lag adjusted t -statistics, in the regression specifications: $R_{i,t} = a + b\Delta S_t + cMKT_t + e_{i,t}$. Excess returns ($R_{i,t}$) are in percentages, and change in sentiment is scaled to have a zero mean and unit standard deviation. Point estimates of b are presented for the bottom decile portfolio (Low beta), the top decile portfolio (High beta), and their differences (H-L). Preranking betas of MOM, PEAD, and PERF are estimated from a single characteristics-based factor model using daily returns over the past 3 months with a minimum of 45 days. Other preranking betas are estimated from a single factor model using monthly returns over the past 60 months with a minimum of 36 months. All portfolios are rebalanced monthly. Ave (Ave_{wo}/o SMB) refers to the average portfolios of all beta-sorted portfolios (excluding SMB). The sample period is from 1970:1 to 2023:12 for IA and ROE, 1975:7 to 2023:12 for FIN and PEAD, and 1966:7 to 2023:12 for others.

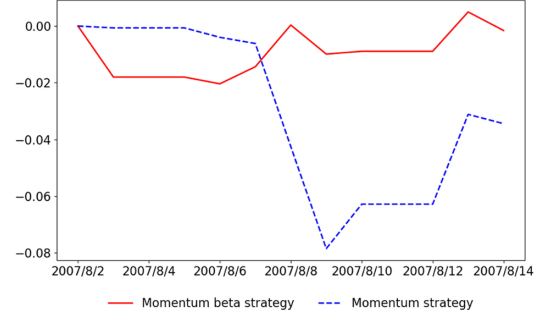


Figure 1
Quant crisis for momentum strategy and momentum beta strategy

This figure presents the cumulative returns for a momentum strategy and a momentum beta strategy during the August 2007 quant crisis. The momentum strategy is a value-weighted portfolio that longs the winner decile of stocks with the highest past 12-month cumulative returns and shorts the loser decile of stocks with the lowest past 12-month cumulative returns, skipping 1 month from the portfolio formation month. The momentum beta strategy is a value-weighted portfolio that longs the top decile of stocks with the largest momentum factor betas and shorts the bottom decile of stocks with the lowest momentum factor betas, where the betas are estimated with respect to the momentum factor using daily returns over the past 3 months with a minimum of 45 days.

5.6 Figure 1: quant crisis and why the beta strategy is different from the anomaly strategy

Read: Figure 1 compares the traditional momentum strategy with the momentum-beta strategy during the August 2007 quant crisis.

- The traditional momentum strategy suffers a sharp drawdown, around 8% at the trough.
- The momentum-beta strategy is far more resilient and remains close to flat, even slightly positive at points.

Why it matters. This figure visually reinforces the point that beta-sorted portfolios are not the same thing as anomaly portfolios. Even when they are based on the same factor family, they load on different stocks and can behave very differently in stress episodes.

5.7 Table 7 and Table 8: mutual funds and hedge funds

Read: These tables ask whether the stock-level result survives when the testing assets are professionally managed portfolios.

- **Mutual funds:** The unconditional long-short spread is only about 0.03% per month, essentially zero. But excluding SMB, it is about -0.32% after low sentiment and $+0.25\%$ after high sentiment, for a difference of about 0.57% per month.
- **Hedge funds:** The unconditional long-short spread is only about 0.08% per month. But excluding SMB, it is about -0.75% after low sentiment and $+0.82\%$ after high sentiment, for a

Table 7
Beta-sorted equity mutual fund portfolio returns following high and low sentiment

	Low beta			High beta			H-L		
	Low Sent.	High Sent.	High - Low	Low Sent.	High Sent.	High - Low	Low Sent.	High Sent.	High - Low
SMB	0.89 (3.28)	0.57 (2.49)	-0.32 (-0.98)	1.39 (3.30)	0.39 (1.09)	-1.00 (-1.85)	0.49 (2.21)	-0.18 (-0.77)	-0.67 (-2.05)
HML	1.41 (3.41)	0.31 (0.83)	-1.10 (-2.10)	0.96 (2.82)	0.89 (3.02)	-0.07 (-0.16)	-0.45 (-1.52)	0.58 (1.63)	1.03 (2.33)
RMW	1.31 (3.41)	0.29 (0.86)	-1.02 (-2.09)	0.96 (2.86)	0.64 (2.41)	-0.33 (-0.80)	-0.34 (-1.37)	0.35 (1.50)	0.69 (2.10)
CMA	1.47 (3.53)	0.37 (0.99)	-1.10 (-2.06)	0.86 (2.80)	0.85 (3.03)	-0.01 (-0.02)	-0.61 (-2.16)	0.48 (1.37)	1.09 (2.52)
IA	1.43 (3.49)	0.38 (1.01)	-1.05 (-2.00)	0.90 (2.83)	0.62 (2.72)	-0.29 (-0.77)	-0.52 (-1.83)	0.24 (0.86)	0.76 (2.01)
ROE	1.08 (2.97)	0.64 (1.77)	-0.44 (-0.88)	1.17 (3.48)	0.63 (2.41)	-0.53 (-1.30)	0.09 (0.37)	-0.01 (-0.02)	-0.09 (-0.24)
PMU	1.02 (2.83)	0.24 (0.80)	-0.79 (-1.76)	1.30 (3.62)	0.71 (2.32)	-0.59 (-1.31)	0.27 (1.07)	0.47 (1.98)	0.20 (0.60)
QMJ	1.35 (3.22)	0.54 (1.36)	-0.81 (-1.44)	0.86 (3.28)	0.59 (2.69)	-0.27 (-0.84)	-0.49 (-2.25)	0.05 (0.16)	0.54 (1.41)
MOM	1.13 (2.55)	0.51 (0.80)	-0.62 (-0.82)	1.17 (2.66)	0.40 (0.91)	-0.76 (-1.37)	0.04 (0.12)	-0.11 (-0.17)	-0.15 (-0.21)
MGMT	1.52 (3.55)	0.36 (0.96)	-1.16 (-2.11)	0.78 (2.69)	0.61 (2.65)	-0.16 (-0.45)	-0.75 (-2.59)	0.25 (0.88)	1.00 (2.54)
PERF	1.18 (2.59)	0.13 (0.19)	-1.05 (-1.29)	1.09 (2.85)	0.57 (1.63)	-0.52 (-1.12)	-0.09 (-0.35)	0.44 (0.72)	0.54 (0.82)
FIN	1.44 (3.39)	0.32 (0.83)	-1.12 (-2.07)	0.87 (3.03)	0.64 (2.95)	-0.23 (-0.65)	-0.57 (-2.00)	0.33 (1.15)	0.90 (2.35)
PEAD	1.32 (2.89)	0.59 (0.99)	-0.72 (-0.98)	1.01 (2.49)	0.30 (0.62)	-0.72 (-1.27)	-0.30 (-0.97)	-0.30 (-0.46)	0.01 (0.01)
Ave	1.28 (3.46)	0.43 (1.33)	-0.85 (-1.80)	1.03 (3.32)	0.65 (2.77)	-0.38 (-1.03)	-0.25 (-2.34)	0.22 (1.51)	0.47 (2.66)
$Ave_{w/o\ SMB}^{CAPM}$	1.31 (3.45)	0.42 (1.24)	-0.90 (-1.83)	0.99 (3.31)	0.67 (2.97)	-0.32 (-0.91)	-0.32 (-2.43)	0.25 (1.48)	0.57 (2.70)
$Ave_{w/o\ SMB}^{CAPM}$	0.08 (0.79)	-0.10 (-0.72)	-0.18 (-1.06)	0.01 (0.16)	0.24 (2.69)	0.23 (2.27)	-0.07 (-0.82)	0.34 (2.47)	0.41 (2.54)
$Ave_{w/o\ SMB}^{CAPM}$	0.09 (0.80)	-0.13 (-0.84)	-0.22 (-1.16)	0.00 (0.04)	0.27 (3.04)	0.27 (2.70)	-0.09 (-0.81)	0.40 (2.53)	0.49 (2.58)

This table reports the results of factor-beta-sorted decile portfolios of actively managed equity mutual funds following high- and low-sentiment regimes, as classified based on the median level of the BW sentiment index. We report the monthly excess returns for the bottom decile portfolio (Low beta), the top decile portfolio (High beta), and their differences (H-L). Preranking betas of MOM, PEAD, and PERF are estimated from a single characteristics-based factor model using daily mutual fund returns over the past 3 months with a minimum of 45 days. Other preranking betas are estimated from a single factor model using monthly mutual fund returns over the past 24 months with a minimum of 18 months. All portfolios are equally weighted and rebalanced monthly. Ave ($Ave_{w/o\ SMB}$) refers to the average portfolios of all beta-sorted portfolios (excluding SMB). We report CAPM alphas for the two average portfolios ($Ave_{w/o\ SMB}^{CAPM}$ and $Ave_{w/o\ SMB}^{CAPM}$) in the last four rows. The sample period is from 1998:12 to 2023:12 for MOM, PERF, and PEAD beta-sorted portfolios, and 1980:1 to 2023:12 for other beta-sorted portfolios. Returns and alphas are in percentages. Newey-West (1987) five-lag adjusted t -statistics are reported in parentheses.

Table 8
Beta-sorted hedge fund portfolio returns following high and low sentiment

	Low beta			High beta			H-L		
	Low Sent.	High Sent.	High - Low	Low Sent.	High Sent.	High - Low	Low Sent.	High Sent.	High - Low
SMB	0.09 (0.50)	0.22 (1.70)	0.13 (0.74)	1.17 (2.91)	0.23 (0.57)	-0.94 (-1.75)	1.08 (3.42)	0.01 (0.02)	-1.07 (-2.09)
HML	0.52 (1.37)	-0.08 (-0.24)	-0.60 (-1.40)	0.80 (2.71)	2.29 (1.38)	1.49 (0.86)	0.28 (0.72)	2.37 (1.57)	2.09 (1.25)
RMW	1.12 (2.33)	-0.15 (-0.41)	-1.27 (-2.38)	0.11 (0.99)	0.59 (2.53)	0.48 (1.90)	-1.01 (-2.13)	0.74 (1.82)	1.75 (3.05)
CMA	0.96 (2.17)	0.15 (0.40)	-0.80 (-1.52)	0.39 (2.03)	2.04 (1.23)	1.65 (0.96)	-0.57 (-1.35)	1.88 (1.22)	2.46 (1.45)
IA	0.97 (2.29)	0.14 (0.35)	-0.82 (-1.57)	0.39 (2.06)	2.11 (1.26)	1.72 (0.99)	-0.58 (-1.52)	1.96 (1.27)	2.55 (1.50)
ROE	1.17 (2.49)	-0.01 (-0.02)	-1.18 (-2.05)	0.19 (1.65)	0.44 (3.36)	0.24 (1.43)	-0.97 (-2.14)	0.45 (1.06)	1.42 (2.48)
PMU	0.94 (2.03)	1.98 (1.11)	1.04 (0.57)	0.39 (1.96)	0.30 (1.36)	-0.09 (-0.38)	-0.55 (-1.14)	-1.68 (-1.02)	-1.13 (-0.64)
QMJ	1.14 (2.28)	0.09 (0.21)	-1.05 (-1.70)	0.12 (1.36)	0.43 (3.75)	0.31 (2.19)	-1.02 (-2.00)	0.34 (0.69)	1.36 (2.08)
MOM	1.18 (3.05)	0.40 (1.16)	-0.78 (-1.60)	0.15 (0.64)	0.08 (0.30)	-0.07 (-0.22)	-1.03 (-3.17)	-0.32 (-0.83)	0.71 (1.44)
MGMT	0.94 (1.95)	0.14 (0.32)	-0.80 (-1.39)	0.32 (2.26)	2.08 (1.27)	1.76 (1.04)	-0.62 (-1.32)	1.94 (1.28)	2.56 (1.52)
PERF	1.22 (3.07)	-0.05 (-0.12)	-1.27 (-2.38)	-0.01 (-0.06)	0.37 (1.92)	0.38 (1.65)	-1.23 (-3.66)	0.42 (1.12)	1.65 (3.30)
FIN	1.14 (2.31)	0.04 (0.10)	-1.09 (-1.88)	0.10 (1.11)	2.27 (1.38)	2.17 (1.28)	-1.04 (-2.12)	2.22 (1.47)	3.26 (1.93)
PEAD	1.05 (2.94)	0.53 (1.74)	-0.52 (-1.17)	0.36 (1.26)	0.00 (0.00)	-0.36 (-1.05)	-0.69 (-2.23)	-0.53 (-1.50)	0.16 (0.36)
Ave	0.96 (2.38)	0.26 (0.66)	-0.69 (-1.37)	0.34 (2.43)	1.02 (1.50)	0.67 (0.97)	-0.61 (-2.10)	0.75 (1.65)	1.37 (2.42)
$Ave_{w/o\ SMB}^{CAPM}$	1.03 (2.42)	0.27 (0.62)	-0.76 (-1.40)	0.28 (2.18)	1.08 (1.50)	0.81 (1.09)	-0.75 (-2.23)	0.82 (1.64)	1.57 (2.50)
$Ave_{w/o\ SMB}^{CAPM}$	0.16 (0.75)	0.00 (0.01)	-0.16 (-0.58)	0.13 (1.15)	0.90 (1.46)	0.77 (1.24)	-0.04 (-0.22)	0.90 (2.10)	0.94 (1.84)
$Ave_{w/o\ SMB}^{CAPM}$	0.19 (0.82)	-0.01 (-0.05)	-0.20 (-0.67)	0.11 (1.01)	0.98 (1.48)	0.87 (1.29)	-0.08 (-0.41)	1.00 (2.16)	1.07 (1.93)

This table reports the results of factor-beta-sorted decile portfolios of hedge funds following high- and low-sentiment regimes, as classified based on the median level of the BW sentiment index. We report the monthly excess returns for the bottom decile portfolio (Low beta), the top decile portfolio (High beta), and their differences (H-L). Preranking betas are estimated from a single factor model using monthly hedge fund returns over the past 24 months with a minimum of 18 months. All portfolios are equally weighted and rebalanced monthly. Ave ($Ave_{w/o\ SMB}$) refers to the average portfolios of all beta-sorted portfolios (excluding SMB). We report CAPM alphas for the two average portfolios ($Ave_{w/o\ SMB}^{CAPM}$ and $Ave_{w/o\ SMB}^{CAPM}$) in the last four rows. Returns and alphas are in percentages. The sample period is from 1996:7 to 2023:12. Newey-West (1987) five-lag adjusted t -statistics are reported in parentheses.

difference of about 1.57% per month.

Alpha version.

- Mutual funds: the CAPM alpha spread difference across the two regimes is about 0.49% per month, with most of the alpha coming from high-sentiment periods.
- Hedge funds: the CAPM alpha spread is about 1.00% per month after high sentiment and slightly negative after low sentiment.

Why it matters. This is very persuasive because professional investors also display the same conditional pattern. The factor beta seems to capture the same kind of mispricing exposure in managed portfolios as in stocks.

5.8 Table 9: alternative sentiment measures

Table 9
Beta-sorted portfolio returns following high and low sentiment: Alternative sentiment measures

	Low beta			High beta			H-L		
	Low Sent.	High Sent.	High - Low	Low Sent.	High Sent.	High - Low	Low Sent.	High Sent.	High - Low
<i>A. Michigan Consumer Sentiment Index</i>									
Ave	1.26 (2.27)	0.12 (0.28)	-1.14 (-1.61)	0.95 (2.46)	0.55 (2.51)	-0.40 (-0.89)	-0.31 (-0.99)	0.43 (1.55)	0.74 (1.78)
Ave _{w/o SMB}	1.31 (2.24)	0.07 (0.14)	-1.24 (-1.66)	0.87 (2.37)	0.63 (3.08)	-0.25 (-0.58)	-0.43 (-1.20)	0.56 (1.69)	0.99 (2.03)
<i>B. Conference Board Consumer Confidence Index</i>									
Ave	1.14 (2.51)	0.10 (0.22)	-1.04 (-1.65)	0.85 (2.50)	0.57 (2.50)	-0.29 (-0.70)	-0.29 (-1.34)	0.47 (1.54)	0.75 (2.05)
Ave _{w/o SMB}	1.18 (2.48)	0.04 (0.09)	-1.14 (-1.71)	0.80 (2.45)	0.63 (2.98)	-0.17 (-0.43)	-0.38 (-1.49)	0.59 (1.66)	0.97 (2.24)
<i>C. PLS sentiment index</i>									
Ave	1.31 (3.70)	-0.44 (-0.96)	-1.75 (-3.15)	0.98 (4.24)	0.17 (0.59)	-0.80 (-2.21)	-0.34 (-1.60)	0.61 (2.51)	0.95 (3.07)
Ave _{w/o SMB}	1.36 (3.62)	-0.50 (-1.04)	-1.86 (-3.16)	0.95 (4.36)	0.23 (0.84)	-0.72 (-2.10)	-0.41 (-1.62)	0.74 (2.53)	1.14 (3.12)
<i>D. AAI Sentiment Survey Index</i>									
Ave	1.40 (2.47)	-0.14 (-0.25)	-1.54 (-1.93)	0.82 (2.28)	0.46 (1.76)	-0.36 (-0.80)	-0.58 (-1.78)	0.60 (1.59)	1.18 (2.37)
Ave _{w/o SMB}	1.46 (2.43)	-0.20 (-0.34)	-1.66 (-1.96)	0.76 (2.20)	0.54 (2.25)	-0.22 (-0.51)	-0.70 (-1.83)	0.74 (1.64)	1.44 (2.43)

This table reports the results of factor-beta-sorted decile value-weighted portfolios following high- and low-sentiment regimes, as classified based on the median level of the Michigan Consumer Sentiment Index (panel A), the Conference Board Consumer Confidence Index (panel B), the Huang et al. (2015) PLS sentiment index (panel C), or the American Association of Individual Investors (AAII) Sentiment Survey Index (panel D). We report the monthly excess returns for the bottom decile portfolio (Low beta), the top decile portfolio (High beta), and their differences (H-L). Preranking betas of MOM, PEAD, and PERF are estimated from a single characteristics-based factor model using daily returns over the past 3 months with a minimum of 45 days. Other preranking betas are estimated from a single-factor model using monthly returns over the past 60 months with a minimum of 36 months. All portfolios are rebalanced monthly. The sample period is from 1970:1 to 2023:12 for IA and ROE, 1975:7 to 2023:12 for FIN and PEAD, and 1966:7 to 2023:12 for others. Ave (Ave_{w/o SMB}) refers to the average portfolios of all beta-sorted portfolios (excluding SMB). We only report these two average portfolios in each panel to save space. The Michigan Consumer Sentiment Index starts from 1978:1, the Conference Board Consumer Confidence Index starts from 1977:6, the PLS sentiment index starts from 1965:7, and the AAI Sentiment Survey Index starts from 1987:7. Returns are in percentages. Newey-West (1987) five-lag adjusted *t*-statistics are reported in parentheses.

Table 10
Beta-sorted portfolios: Predictive regressions on lagged sentiment and additional macro variables

	Low beta		High beta		H-L	
	$\hat{b}$	<i>t</i> -stat.	$\hat{b}$	<i>t</i> -stat.	$\hat{b}$	<i>t</i> -stat.
SMB	-0.35	(-2.09)	-1.20	(-2.78)	-0.84	(-2.14)
HML	-1.43	(-3.27)	-0.10	(-0.36)	1.33	(2.76)
RMW	-1.63	(-3.29)	-0.30	(-1.42)	1.33	(2.89)
CMA	-1.27	(-3.05)	0.01	(0.04)	1.28	(2.93)
IA	-1.38	(-3.15)	-0.03	(-0.10)	1.36	(2.82)
ROE	-1.30	(-2.72)	-0.37	(-1.87)	0.93	(2.26)
PMU	-0.58	(-1.60)	-0.81	(-2.55)	-0.24	(-0.59)
QMJ	-1.33	(-2.87)	-0.23	(-1.36)	1.09	(2.58)
MOM	-0.94	(-2.49)	-0.48	(-1.55)	0.46	(1.17)
MGMT	-1.57	(-3.24)	-0.10	(-0.44)	1.47	(3.08)
PERF	-1.68	(-3.71)	-0.41	(-1.72)	1.27	(3.21)
FIN	-1.65	(-3.43)	-0.25	(-1.35)	1.39	(3.01)
PEAD	-0.78	(-1.99)	-1.08	(-2.77)	-0.29	(-0.55)
Ave	-1.22	(-3.41)	-0.41	(-2.01)	0.81	(3.39)
Ave _{w/o SMB}	-1.29	(-3.41)	-0.35	(-1.76)	0.95	(3.40)

This table reports the point estimates of $\hat{b}$ ($\hat{b}$), along with Newey-West (1987) five-lag adjusted *t*-statistics, in the regression specifications: $R_{i,t} = a + b \times S_{i,t-1} + \sum_{j=1}^4 m_j \times X_{j,t-1} + c \times TED_{t-1} + d \times Inf_{t-1} + e \times Disp_{t-1} + e_{i,t}$. Macro variables $X_{j,t-1}$ include the default premium, the term premium, the real interest rate, and the log wealth-consumption ratio. Additional control variables include the difference between the 3-month LIBOR and Treasury-bill rate (TED), the inflation rate, and the beta-weighted aggregate dispersion (Disp) as in Hong and Sraer (2016). Excess returns ($R_{i,t}$) are in percentages. Point estimates of $\hat{b}$ are presented for the bottom decile portfolio (Low beta), the top decile portfolio (High beta), and their differences (H-L). Preranking betas of MOM, PEAD, and PERF are estimated from a single characteristics-based factor model using daily returns over the past 3 months with a minimum of 45 days. Other preranking betas are estimated from a single-factor model using monthly returns over the past 60 months with a minimum of 36 months. All portfolios are rebalanced monthly. Ave (Ave_{w/o SMB}) refers to the average portfolios of all beta-sorted portfolios (excluding SMB). The sample period is from 1986:2 to 2023:12.

Read: This table shows the result is not an artifact of the BW index.

For the average portfolio excluding SMB, the difference in high-minus-low spreads across sentiment regimes is approximately:

- 0.99% using the Michigan Consumer Sentiment Index,
- 0.97% using the Conference Board index,
- 1.14% using the Huang et al. PLS sentiment index,
- 1.44% using the AAI sentiment survey.

Why it matters. The sign is stable, and the magnitude is often even larger than with BW.

5.9 Table 10, Table 11, and Table 12: alternative mechanisms and placebo tests

Read: These tables ask whether sentiment is just standing in for something else.

Table 11
Beta-sorted portfolios: after high and low state

A. After low-state period															
	dp	dy	op	de	ovar	bms	mtis	tbl	by	lr	tms	dty	dfr	infl	sup
Ave	0.21 (0.58)	0.20 (1.46)	0.24 (2.16)	0.16 (0.72)	0.04 (0.13)	0.22 (1.34)	0.11 (0.52)	-0.04 (-0.16)	0.06 (0.22)	0.03 (0.15)	0.24 (1.14)	0.02 (0.09)	-0.03 (-0.12)	-0.06 (-0.30)	0.17 (0.48)
Ans _{10/10} SMB	0.22 (1.35)	0.21 (1.22)	0.27 (1.85)	0.19 (0.71)	0.00 (0.00)	0.24 (1.32)	0.13 (0.50)	-0.10 (-0.34)	0.07 (0.23)	0.04 (0.19)	0.26 (1.06)	-0.04 (-0.10)	-0.03 (-0.11)	-0.08 (-0.32)	0.23 (0.38)
B. After high-state period															
Ave	0.09 (0.33)	0.10 (0.37)	0.05 (0.19)	0.14 (0.67)	0.26 (2.20)	0.08 (0.30)	0.18 (0.81)	0.34 (1.77)	0.24 (1.51)	0.26 (1.40)	0.06 (0.27)	0.19 (1.15)	0.33 (1.54)	0.37 (2.09)	0.40 (1.79)
Ans _{10/10} SMB	0.13 (0.41)	0.15 (0.46)	0.09 (0.27)	0.16 (0.67)	0.36 (2.41)	0.12 (0.37)	0.22 (0.82)	0.45 (1.99)	0.39 (1.43)	0.45 (1.36)	0.07 (0.34)	0.09 (0.25)	0.24 (1.51)	0.39 (2.12)	0.45 (1.65)
C. High-low															
Ave	-0.12 (-0.41)	-0.10 (-0.33)	-0.19 (-0.60)	-0.02 (0.14)	0.22 (-0.46)	-0.14 (0.23)	0.07 (0.19)	0.38 (0.63)	0.18 (0.03)	0.23 (-0.61)	0.17 (0.40)	0.35 (1.21)	0.43 (1.76)	0.49 (1.69)	-0.10 (-0.26)
Ans _{10/10} SMB	-0.10 (-0.27)	-0.06 (-0.17)	-0.18 (-0.50)	0.36 (1.00)	-0.12 (-0.34)	0.10 (0.27)	0.56 (1.48)	0.22 (0.63)	0.26 (0.94)	-0.17 (-0.49)	0.29 (0.58)	0.42 (1.29)	0.53 (1.81)	0.54 (1.52)	-0.16 (-0.35)

This table reports the results of factor beta-sorted value-weighted decile portfolios following high- and low-state regimes, as classified based on the median level of the 14 equity premium predictors as in Welch and Goyal (2008), the surplus ratio, and the aggregate dispersion as in Hong and Sier (2016). The 14 predictors include the log dividend-price ratio (dp), log dividend yield (dy), log earnings-price ratio (op), log dividend-payout ratio (de), stock variance (ovar), book-to-market ratio (bms), net equity issuance (mtis), Treasury-bill rate (tbl), long-term yield (by), long-term return (lr), term spread (tms), default yield spread (dty), default return spread (dfr), and inflation (infl). We take the negative of dp, dy, op, de, ovar, bms, lr, tms, dty, and dfr to be consistent with other predictors that a high value predicts a lower return. We report the monthly excess returns of the high-low beta average portfolios after low-state periods (panel A), high-state periods (panel B), and their differences (panel C). Premaking betas of MOM, PREAD, and PDEF are estimated from a single characteristics-based factor model using daily returns over the past 3 months with a minimum of 45 days. Other premaking betas are estimated from a single-factor model using monthly returns over the past 60 months with a minimum of 18 months. All portfolios are rebalanced monthly. Ave (Ans_{10/10}) refers to the average portfolios of all beta-sorted portfolios (excluding SMB). The sample period is from 1982:1 to 2023:12 for the specification with aggregate dispersion as the control variable and 1966:7 to 2023:12 for other specifications. Returns are in percentages. Newey-West (1987) five-lag adjusted *t*-statistics are reported in parentheses.

Table 12
Beta-sorted portfolios: Predictive regressions on lagged sentiment controlling for additional state variables

A. Ave															
Low beta				High beta				H-L				B. Ans _{10/10} SMB			
$\hat{\beta}$	<i>t</i> -stat.	$\hat{\beta}$	<i>t</i> -stat.	$\hat{\beta}$	<i>t</i> -stat.	$\hat{\beta}$	<i>t</i> -stat.	$\hat{\beta}$	<i>t</i> -stat.	$\hat{\beta}$	<i>t</i> -stat.	$\hat{\beta}$	<i>t</i> -stat.	$\hat{\beta}$	<i>t</i> -stat.
dp	-0.80 (-2.57)	-0.26 (-1.10)	0.54 (3.75)	-0.26 (-1.05)	0.53 (3.73)	-0.86 (-2.66)	-0.20 (-0.87)	0.67 (3.93)	-0.86 (-2.66)	-0.20 (-0.87)	0.67 (3.93)	-0.86 (-2.66)	-0.20 (-0.87)	0.67 (3.93)	-0.86 (-2.66)
dy	-0.79 (-2.52)	-0.25 (-1.05)	0.53 (3.73)	-0.79 (-2.52)	-0.25 (-1.05)	0.53 (3.73)	-0.79 (-2.52)	-0.25 (-1.05)	0.53 (3.73)	-0.79 (-2.52)	-0.25 (-1.05)	0.53 (3.73)	-0.79 (-2.52)	-0.25 (-1.05)	0.53 (3.73)
op	-0.88 (-2.78)	-0.28 (-1.17)	0.60 (4.02)	-0.88 (-2.78)	-0.28 (-1.17)	0.60 (4.02)	-0.88 (-2.78)	-0.28 (-1.17)	0.60 (4.02)	-0.88 (-2.78)	-0.28 (-1.17)	0.60 (4.02)	-0.88 (-2.78)	-0.28 (-1.17)	0.60 (4.02)
de	-0.78 (-2.52)	-0.27 (-1.20)	0.50 (3.63)	-0.78 (-2.52)	-0.27 (-1.20)	0.50 (3.63)	-0.78 (-2.52)	-0.27 (-1.20)	0.50 (3.63)	-0.78 (-2.52)	-0.27 (-1.20)	0.50 (3.63)	-0.78 (-2.52)	-0.27 (-1.20)	0.50 (3.63)
ovar	-0.78 (-2.50)	-0.28 (-1.21)	0.51 (3.45)	-0.78 (-2.50)	-0.28 (-1.21)	0.51 (3.45)	-0.78 (-2.50)	-0.28 (-1.21)	0.51 (3.45)	-0.78 (-2.50)	-0.28 (-1.21)	0.51 (3.45)	-0.78 (-2.50)	-0.28 (-1.21)	0.51 (3.45)
bms	-0.90 (-2.86)	-0.29 (-1.17)	0.61 (4.01)	-0.90 (-2.86)	-0.29 (-1.17)	0.61 (4.01)	-0.90 (-2.86)	-0.29 (-1.17)	0.61 (4.01)	-0.90 (-2.86)	-0.29 (-1.17)	0.61 (4.01)	-0.90 (-2.86)	-0.29 (-1.17)	0.61 (4.01)
mtis	-0.79 (-2.54)	-0.28 (-1.23)	0.51 (3.46)	-0.79 (-2.54)	-0.28 (-1.23)	0.51 (3.46)	-0.79 (-2.54)	-0.28 (-1.23)	0.51 (3.46)	-0.79 (-2.54)	-0.28 (-1.23)	0.51 (3.46)	-0.79 (-2.54)	-0.28 (-1.23)	0.51 (3.46)
tbl	-0.68 (-2.34)	-0.18 (-0.82)	0.50 (3.40)	-0.68 (-2.34)	-0.18 (-0.82)	0.50 (3.40)	-0.68 (-2.34)	-0.18 (-0.82)	0.50 (3.40)	-0.68 (-2.34)	-0.18 (-0.82)	0.50 (3.40)	-0.68 (-2.34)	-0.18 (-0.82)	0.50 (3.40)
by	-0.77 (-2.48)	-0.27 (-1.19)	0.50 (3.39)	-0.77 (-2.48)	-0.27 (-1.19)	0.50 (3.39)	-0.77 (-2.48)	-0.27 (-1.19)	0.50 (3.39)	-0.77 (-2.48)	-0.27 (-1.19)	0.50 (3.39)	-0.77 (-2.48)	-0.27 (-1.19)	0.50 (3.39)
lr	-0.79 (-2.55)	-0.28 (-1.24)	0.51 (3.47)	-0.79 (-2.55)	-0.28 (-1.24)	0.51 (3.47)	-0.79 (-2.55)	-0.28 (-1.24)	0.51 (3.47)	-0.79 (-2.55)	-0.28 (-1.24)	0.51 (3.47)	-0.79 (-2.55)	-0.28 (-1.24)	0.51 (3.47)
tms	-0.77 (-2.49)	-0.26 (-1.16)	0.51 (3.48)	-0.77 (-2.49)	-0.26 (-1.16)	0.51 (3.48)	-0.77 (-2.49)	-0.26 (-1.16)	0.51 (3.48)	-0.77 (-2.49)	-0.26 (-1.16)	0.51 (3.48)	-0.77 (-2.49)	-0.26 (-1.16)	0.51 (3.48)
dty	-0.75 (-2.44)	-0.25 (-1.12)	0.50 (3.42)	-0.75 (-2.44)	-0.25 (-1.12)	0.50 (3.42)	-0.75 (-2.44)	-0.25 (-1.12)	0.50 (3.42)	-0.75 (-2.44)	-0.25 (-1.12)	0.50 (3.42)	-0.75 (-2.44)	-0.25 (-1.12)	0.50 (3.42)
dfr	-0.78 (-2.47)	-0.27 (-1.18)	0.51 (3.44)	-0.78 (-2.47)	-0.27 (-1.18)	0.51 (3.44)	-0.78 (-2.47)	-0.27 (-1.18)	0.51 (3.44)	-0.78 (-2.47)	-0.27 (-1.18)	0.51 (3.44)	-0.78 (-2.47)	-0.27 (-1.18)	0.51 (3.44)
infl	-0.83 (-2.76)	-0.29 (-1.27)	0.54 (3.71)	-0.83 (-2.76)	-0.29 (-1.27)	0.54 (3.71)	-0.83 (-2.76)	-0.29 (-1.27)	0.54 (3.71)	-0.83 (-2.76)	-0.29 (-1.27)	0.54 (3.71)	-0.83 (-2.76)	-0.29 (-1.27)	0.54 (3.71)
supl	-0.85 (-2.94)	-0.31 (-1.51)	0.53 (3.64)	-0.85 (-2.94)	-0.31 (-1.51)	0.53 (3.64)	-0.85 (-2.94)	-0.31 (-1.51)	0.53 (3.64)	-0.85 (-2.94)	-0.31 (-1.51)	0.53 (3.64)	-0.85 (-2.94)	-0.31 (-1.51)	0.53 (3.64)
dmp	-1.19 (-3.85)	-0.33 (-1.66)	0.86 (4.56)	-1.19 (-3.85)	-0.33 (-1.66)	0.86 (4.56)	-1.19 (-3.85)	-0.33 (-1.66)	0.86 (4.56)	-1.19 (-3.85)	-0.33 (-1.66)	0.86 (4.56)	-1.19 (-3.85)	-0.33 (-1.66)	0.86 (4.56)

This table reports the point estimates of $\hat{\beta}$ ($\hat{\beta}$), along with Newey-West (1987) five-lag adjusted *t*-statistics, in the regression specification: $R_{i,t} = \alpha + \beta_1 \Delta \ln D_{i,t} + \beta_2 \Delta \ln Y_{i,t} + \beta_3 \Delta \ln E_{i,t} + \beta_4 \Delta \ln P_{i,t} + \beta_5 \Delta \ln V_{i,t} + \beta_6 \Delta \ln B_{i,t} + \beta_7 \Delta \ln M_{i,t} + \beta_8 \Delta \ln T_{i,t} + \beta_9 \Delta \ln I_{i,t} + \beta_{10} \Delta \ln S_{i,t} + \beta_{11} \Delta \ln R_{i,t} + \beta_{12} \Delta \ln C_{i,t} + \beta_{13} \Delta \ln F_{i,t} + \beta_{14} \Delta \ln G_{i,t} + \beta_{15} \Delta \ln H_{i,t} + \beta_{16} \Delta \ln J_{i,t} + \beta_{17} \Delta \ln K_{i,t} + \beta_{18} \Delta \ln L_{i,t} + \beta_{19} \Delta \ln M_{i,t} + \beta_{20} \Delta \ln N_{i,t} + \beta_{21} \Delta \ln O_{i,t} + \beta_{22} \Delta \ln P_{i,t} + \beta_{23} \Delta \ln Q_{i,t} + \beta_{24} \Delta \ln R_{i,t} + \beta_{25} \Delta \ln S_{i,t} + \beta_{26} \Delta \ln T_{i,t} + \beta_{27} \Delta \ln U_{i,t} + \beta_{28} \Delta \ln V_{i,t} + \beta_{29} \Delta \ln W_{i,t} + \beta_{30} \Delta \ln X_{i,t} + \beta_{31} \Delta \ln Y_{i,t} + \beta_{32} \Delta \ln Z_{i,t} + \beta_{33} \Delta \ln A_{i,t} + \beta_{34} \Delta \ln B_{i,t} + \beta_{35} \Delta \ln C_{i,t} + \beta_{36} \Delta \ln D_{i,t} + \beta_{37} \Delta \ln E_{i,t} + \beta_{38} \Delta \ln F_{i,t} + \beta_{39} \Delta \ln G_{i,t} + \beta_{40} \Delta \ln H_{i,t} + \beta_{41} \Delta \ln I_{i,t} + \beta_{42} \Delta \ln J_{i,t} + \beta_{43} \Delta \ln K_{i,t} + \beta_{44} \Delta \ln L_{i,t} + \beta_{45} \Delta \ln M_{i,t} + \beta_{46} \Delta \ln N_{i,t} + \beta_{47} \Delta \ln O_{i,t} + \beta_{48} \Delta \ln P_{i,t} + \beta_{49} \Delta \ln Q_{i,t} + \beta_{50} \Delta \ln R_{i,t} + \beta_{51} \Delta \ln S_{i,t} + \beta_{52} \Delta \ln T_{i,t} + \beta_{53} \Delta \ln U_{i,t} + \beta_{54} \Delta \ln V_{i,t} + \beta_{55} \Delta \ln W_{i,t} + \beta_{56} \Delta \ln X_{i,t} + \beta_{57} \Delta \ln Y_{i,t} + \beta_{58} \Delta \ln Z_{i,t} + \beta_{59} \Delta \ln A_{i,t} + \beta_{60} \Delta \ln B_{i,t} + \beta_{61} \Delta \ln C_{i,t} + \beta_{62} \Delta \ln D_{i,t} + \beta_{63} \Delta \ln E_{i,t} + \beta_{64} \Delta \ln F_{i,t} + \beta_{65} \Delta \ln G_{i,t} + \beta_{66} \Delta \ln H_{i,t} + \beta_{67} \Delta \ln I_{i,t} + \beta_{68} \Delta \ln J_{i,t} + \beta_{69} \Delta \ln K_{i,t} + \beta_{70} \Delta \ln L_{i,t} + \beta_{71} \Delta \ln M_{i,t} + \beta_{72} \Delta \ln N_{i,t} + \beta_{73} \Delta \ln O_{i,t} + \beta_{74} \Delta \ln P_{i,t} + \beta_{75} \Delta \ln Q_{i,t} + \beta_{76} \Delta \ln R_{i,t} + \beta_{77} \Delta \ln S_{i,t} + \beta_{78} \Delta \ln T_{i,t} + \beta_{79} \Delta \ln U_{i,t} + \beta_{80} \Delta \ln V_{i,t} + \beta_{81} \Delta \ln W_{i,t} + \beta_{82} \Delta \ln X_{i,t} + \beta_{83} \Delta \ln Y_{i,t} + \beta_{84} \Delta \ln Z_{i,t} + \beta_{85} \Delta \ln A_{i,t} + \beta_{86} \Delta \ln B_{i,t} + \beta_{87} \Delta \ln C_{i,t} + \beta_{88} \Delta \ln D_{i,t} + \beta_{89} \Delta \ln E_{i,t} + \beta_{90} \Delta \ln F_{i,t} + \beta_{91} \Delta \ln G_{i,t} + \beta_{92} \Delta \ln H_{i,t} + \beta_{93} \Delta \ln I_{i,t} + \beta_{94} \Delta \ln J_{i,t} + \beta_{95} \Delta \ln K_{i,t} + \beta_{96} \Delta \ln L_{i,t} + \beta_{97} \Delta \ln M_{i,t} + \beta_{98} \Delta \ln N_{i,t} + \beta_{99} \Delta \ln O_{i,t} + \beta_{100} \Delta \ln P_{i,t} + \beta_{101} \Delta \ln Q_{i,t} + \beta_{102} \Delta \ln R_{i,t} + \beta_{103} \Delta \ln S_{i,t} + \beta_{104} \Delta \ln T_{i,t} + \beta_{105} \Delta \ln U_{i,t} + \beta_{106} \Delta \ln V_{i,t} + \beta_{107} \Delta \ln W_{i,t} + \beta_{108} \Delta \ln X_{i,t} + \beta_{109} \Delta \ln Y_{i,t} + \beta_{110} \Delta \ln Z_{i,t} + \beta_{111} \Delta \ln A_{i,t} + \beta_{112} \Delta \ln B_{i,t} + \beta_{113} \Delta \ln C_{i,t} + \beta_{114} \Delta \ln D_{i,t} + \beta_{115} \Delta \ln E_{i,t} + \beta_{116} \Delta \ln F_{i,t} + \beta_{117} \Delta \ln G_{i,t} + \beta_{118} \Delta \ln H_{i,t} + \beta_{119} \Delta \ln I_{i,t} + \beta_{120} \Delta \ln J_{i,t} + \beta_{121} \Delta \ln K_{i,t} + \beta_{122} \Delta \ln L_{i,t} + \beta_{123} \Delta \ln M_{i,t} + \beta_{124} \Delta \ln N_{i,t} + \beta_{125} \Delta \ln O_{i,t} + \beta_{126} \Delta \ln P_{i,t} + \beta_{127} \Delta \ln Q_{i,t} + \beta_{128} \Delta \ln R_{i,t} + \beta_{129} \Delta \ln S_{i,t} + \beta_{130} \Delta \ln T_{i,t} + \beta_{131} \Delta \ln U_{i,t} + \beta_{132} \Delta \ln V_{i,t} + \beta_{133} \Delta \ln W_{i,t} + \beta_{134} \Delta \ln X_{i,t} + \beta_{135} \Delta \ln Y_{i,t} + \beta_{136} \Delta \ln Z_{i,t} + \beta_{137} \Delta \ln A_{i,t} + \beta_{138} \Delta \ln B_{i,t} + \beta_{139} \Delta \ln C_{i,t} + \beta_{140} \Delta \ln D_{i,t} + \beta_{141} \Delta \ln E_{i,t} + \beta_{142} \Delta \ln F_{i,t} + \beta_{143} \Delta \ln G_{i,t} + \beta_{144} \Delta \ln H_{i,t} + \beta_{145} \Delta \ln I_{i,t} + \beta_{146} \Delta \ln J_{i,t} + \beta_{147} \Delta \ln K_{i,t} + \beta_{148} \Delta \ln L_{i,t} + \beta_{149} \Delta \ln M_{i,t} + \beta_{150} \Delta \ln N_{i,t} + \beta_{151} \Delta \ln O_{i,t} + \beta_{152} \Delta \ln P_{i,t} + \beta_{153} \Delta \ln Q_{i,t} + \beta_{154} \Delta \ln R_{i,t} + \beta_{155} \Delta \ln S_{i,t} + \beta_{156} \Delta \ln T_{i,t} + \beta_{157} \Delta \ln U_{i,t} + \beta_{158} \Delta \ln V_{i,t} + \beta_{159} \Delta \ln W_{i,t} + \beta_{160} \Delta \ln X_{i,t} + \beta_{161} \Delta \ln Y_{i,t} + \beta_{162} \Delta \ln Z_{i,t} + \beta_{163} \Delta \ln A_{i,t} + \beta_{164} \Delta \ln B_{i,t} + \beta_{165} \Delta \ln C_{i,t} + \beta_{166} \Delta \ln D_{i,t} + \beta_{167} \Delta \ln E_{i,t} + \beta_{168} \Delta \ln F_{i,t} + \beta_{169} \Delta \ln G_{i,t} + \beta_{170} \Delta \ln H_{i,t} + \beta_{171} \Delta \ln I_{i,t} + \beta_{172} \Delta \ln J_{i,t} + \beta_{173} \Delta \ln K_{i,t} + \beta_{174} \Delta \ln L_{i,t} + \beta_{175} \Delta \ln M_{i,t} + \beta_{176} \Delta \ln N_{i,t} + \beta_{177} \Delta \ln O_{i,t} + \beta_{178} \Delta \ln P_{i,t} + \beta_{179} \Delta \ln Q_{i,t} + \beta_{180} \Delta \ln R_{i,t} + \beta_{181} \Delta \ln S_{i,t} + \beta_{182} \Delta \ln T_{i,t} + \beta_{183} \Delta \ln U_{i,t} + \beta_{184} \Delta \ln V_{i,t} + \beta_{185} \Delta \ln W_{i,t} + \beta_{186} \Delta \ln X_{i,t} + \beta_{187} \Delta \ln Y_{i,t} + \beta_{188} \Delta \ln Z_{i,t} + \beta_{189} \Delta \ln A_{i,t} + \beta_{190} \Delta \ln B_{i,t} + \beta_{191} \Delta \ln C_{i,t} + \beta_{192} \Delta \ln D_{i,t} + \beta_{193} \Delta \ln E_{i,t} + \beta_{194} \Delta \ln F_{i,t} + \beta_{195} \Delta \ln G_{i,t} + \beta_{196} \Delta \ln H_{i,t} + \beta_{197} \Delta \ln I_{i,t} + \beta_{198} \Delta \ln J_{i,t} + \beta_{199} \Delta \ln K_{i,t} + \beta_{200} \Delta \ln L_{i,t} + \beta_{201} \Delta \ln M_{i,t} + \beta_{202} \Delta \ln N_{i,t} + \beta_{203} \Delta \ln O_{i,t} + \beta_{204} \Delta \ln P_{i,t} + \beta_{205} \Delta \ln Q_{i,t} + \beta_{206} \Delta \ln R_{i,t} + \beta_{207} \Delta \ln S_{i,t} + \beta_{208} \Delta \ln T_{i,t} + \beta_{209} \Delta \ln U_{i,t} + \beta_{210} \Delta \ln V_{i,t} + \beta_{211} \Delta \ln W_{i,t} + \beta_{212} \Delta \ln X_{i,t} + \beta_{213} \Delta \ln Y_{i,t} + \beta_{214} \Delta \ln Z_{i,t} + \beta_{215} \Delta \ln A_{i,t} + \beta_{216} \Delta \ln B_{i,t} + \beta_{217} \Delta \ln C_{i,t} + \beta_{218} \Delta \ln D_{i,t} + \beta_{219} \Delta \ln E_{i,t} + \beta_{220} \Delta \ln F_{i,t} + \beta_{221} \Delta \ln G_{i,t} + \beta_{222} \Delta \ln H_{i,t} + \beta_{223} \Delta \ln I_{i,t} + \beta_{224} \Delta \ln J_{i,t} + \beta_{225} \Delta \ln K_{i,t} + \beta_{226} \Delta \ln L_{i,t} + \beta_{227} \Delta \ln M_{i,t} + \beta_{228} \Delta \ln N_{i,t} + \beta_{229} \Delta \ln O_{i,t} + \beta_{230} \Delta \ln P_{i,t} + \beta_{231} \Delta \ln Q_{i,t} + \beta_{232} \Delta \ln R_{i,t} + \beta_{233} \Delta \ln S_{i,t} + \beta_{234} \Delta \ln T_{i,t} + \beta_{235} \Delta \ln U_{i,t} + \beta_{236} \Delta \ln V_{i,t} + \beta_{237} \Delta \ln W_{i,t} + \beta_{238} \Delta \ln X_{i,t} + \beta_{239} \Delta \ln Y_{i,t} + \beta_{240} \Delta \ln Z_{i,t} + \beta_{241} \Delta \ln A_{i,t} + \beta_{242} \Delta \ln B_{i,t} + \beta_{243} \Delta \ln C_{i,t} + \beta_{244} \Delta \ln D_{i,t} + \beta_{245} \Delta \ln E_{i,t} + \beta_{246} \Delta \ln F_{i,t} + \beta_{247} \Delta \ln G_{i,t} + \beta_{248} \Delta \ln H_{i,t} + \beta_{249} \Delta \ln I_{i,t} + \beta_{250} \Delta \ln J_{i,t} + \beta_{251} \Delta \ln K_{i,t} + \beta_{252} \Delta \ln L_{i,t} + \beta_{253} \Delta \ln M_{i,t} + \beta_{254} \Delta \ln N_{i,t} + \beta_{255} \Delta \ln O_{i,t} + \beta_{256} \Delta \ln P_{i,t} + \beta_{257} \Delta \ln Q_{i,t} + \beta_{258} \Delta \ln R_{i,t} + \beta_{259} \Delta \ln S_{i,t} + \beta_{260} \Delta \ln T_{i,t} + \beta_{261} \Delta \ln U_{i,t} + \beta_{262} \Delta \ln V_{i,t} + \beta_{263} \Delta \ln W_{i,t} + \beta_{264} \Delta \ln X_{i,t} + \beta_{265} \Delta \ln Y_{i,t} + \beta_{266} \Delta \ln Z_{i,t} + \beta_{267} \Delta \ln A_{i,t} + \beta_{268} \Delta \ln B_{i,t} + \beta_{269} \Delta \ln C_{i,t} + \beta_{270} \Delta \ln D_{i,t} + \beta_{271} \Delta \ln E_{i,t} + \beta_{272} \Delta \ln F_{i,t} + \beta_{273} \Delta \ln G_{i,t} + \beta_{274} \Delta \ln H_{i,t} + \beta_{275} \Delta \ln I_{i,t} + \beta_{276} \Delta \ln J_{i,t} + \beta_{277} \Delta \ln K_{i,t} + \beta_{278} \Delta \ln L_{i,t} + \beta_{279} \Delta \ln M_{i,t} + \beta_{280} \Delta \ln N_{i,t} + \beta_{281} \Delta \ln O_{i,t} + \beta_{282} \Delta \ln P_{i,t} + \beta_{283} \Delta$

Table 13
Beta-sorted portfolio overnight and intraday returns

	Low beta			High beta			H-L		
	ON	Intra	Intra-ON	ON	Intra	Intra-ON	ON	Intra	Intra-ON
<i>A. Characteristics-based factor beta-sorted portfolios</i>									
Ave	1.84 (7.20)	-0.48 (-1.45)	-2.33 (-5.46)	1.12 (7.29)	0.04 (0.20)	-1.08 (-4.61)	-0.72 (-4.51)	0.52 (2.18)	1.24 (3.85)
Ave _{w/o SMB}	1.96 (7.24)	-0.57 (-1.60)	-2.53 (-5.56)	1.01 (6.89)	0.11 (0.65)	-0.90 (-3.92)	-0.95 (-4.97)	0.68 (2.41)	1.63 (4.25)
<i>B. Macro-factor beta-sorted portfolios</i>									
Ave1	0.91 (5.52)	0.28 (1.59)	-0.63 (-2.52)	2.00 (7.32)	-0.41 (-1.19)	-2.41 (-5.49)	1.09 (6.77)	-0.69 (-2.66)	-1.78 (-5.41)
Ave2	1.02 (5.44)	0.22 (1.25)	-0.80 (-2.94)	1.94 (7.43)	-0.37 (-1.09)	-2.31 (-5.35)	0.92 (6.85)	-0.60 (-2.46)	-1.52 (-5.02)
Ave3	1.06 (5.47)	0.17 (0.91)	-0.89 (-3.12)	1.90 (7.12)	-0.35 (-1.04)	-2.25 (-5.12)	0.84 (6.47)	-0.52 (-2.27)	-1.36 (-4.62)

This table reports the overnight returns (ON), intraday returns (Intra), and their differences (Intra-ON) of characteristics-factor beta-sorted (panel A) and macro-factor beta-sorted (panel B) decile value-weighted portfolios. We report the bottom decile portfolio (Low beta), the top decile portfolio (High beta), and their differences (H-L). Following Lou, Polk, and Skouras (2019), the intraday return is the price appreciation between market open and close of each day, and the overnight return is imputed based on daily return and this intraday return. Beta-sorted portfolios in panel A are constructed in the same way as in Table 3. In panel A, preranking betas of MOM, PEAD, and PERF are estimated from a single characteristics-based factor model using daily returns over the past 3 months with a minimum of 45 days, and other preranking betas are estimated from a single-factor model using monthly returns over the past 60 months with a minimum of 36 months. Ave (Ave_{w/o SMB}) refers to the average portfolios of all beta-sorted portfolios (excluding SMB). In panel B, we consider 10 macro factors including consumption growth (CON), industrial production growth (IPG), term premium (TERM), default premium (DEF), unexpected inflation (UI), changes in expected inflation (DEI), aggregate market volatility (VOL), market excess returns (MKT), labor income growth (LAB), and TFP growth (TFP). Following Shen, Yu, and Zhao (2017), we take the inverse of the betas based on DEF, TERM, and VOL to be consistent with other macro-factor betas that high beta portfolios have high returns. Ave1 refers to the average portfolios of beta-sorted portfolios including CON, TFP, IPG, VOL, LAB, and MKT; Ave2 adds TERM and DEF beta-sorted portfolios; and Ave3 includes all beta-sorted portfolios. Preranking betas of TFP are estimated from a single-factor model using quarterly data over the past 24 quarters with a minimum of 12 quarters, and other preranking betas are estimated from a single-factor model using monthly returns over the past 60 months with a minimum of 36 months. All portfolios are rebalanced monthly. We only report the average portfolios in each panel to save space. The sample period is from 1992:7 to 2023:12. Returns are in percentages. Newey-West (1987) five-lag adjusted *t*-statistics are reported in parentheses.

- **Core empirical fact.** Beta-sorted portfolios are almost unpriced on average, but exhibit a very strong conditional pattern: high-beta outperforms after high sentiment, while low-beta outperforms after low sentiment.
- **Interpretation.** This sign pattern is consistent with characteristics-based factor beta being a noisy proxy for mispricing, not risk.
- **Cross-sectional mechanism.** Low-beta firms are more sensitive to sentiment changes and appear especially overpriced after high sentiment.
- **Robustness.** The result survives alternative sentiment indices, factor-beta estimation windows, mutual fund and hedge fund tests, additional macro controls, placebo predictors, and overnight/intraday decomposition.
- **Main implication.** Using characteristics-based factors as risk-adjustment tools in anomaly studies, attribution, and event studies can be conceptually misleading if those factors primarily reflect mispricing.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the pricing of characteristics-based factor-beta portfolios is strongly state dependent. The unconditional beta-return relation is weak, but conditional on investor sentiment the pattern is sharp and systematic: high-beta beats low-beta after high sentiment, and the reverse holds after low sentiment. This is the exact opposite of what is found for macro-related factors.

7.2 Economic intuition

The paper’s intuition is very clean:

- high sentiment makes overpricing more likely because short-sale constraints keep rational traders from fully arbitraging it away;
- if factor beta is a noisy proxy for mispricing, then the beta-return relation should strengthen precisely in those high-sentiment states;
- low-beta firms appear to be the ones most affected by sentiment, so the conditional pattern comes largely from the short leg.

Thus, characteristics-based factor beta behaves more like a measure of relative mispricing than a measure of systematic risk.

7.3 Takeaway

The one-sentence takeaway is:

When returns are sorted directly on characteristics-based factor beta, sentiment reveals what the factor means. The fact that high-beta portfolios outperform after *high* sentiment—not after low sentiment—strongly suggests that these betas proxy for mispricing rather than risk.

A second takeaway is methodological:

Do not equate “high factor beta” with “having the underlying characteristic.” The paper shows those are different objects, and that distinction matters for pricing tests, anomaly evaluation, fund attribution, and event-study adjustment.